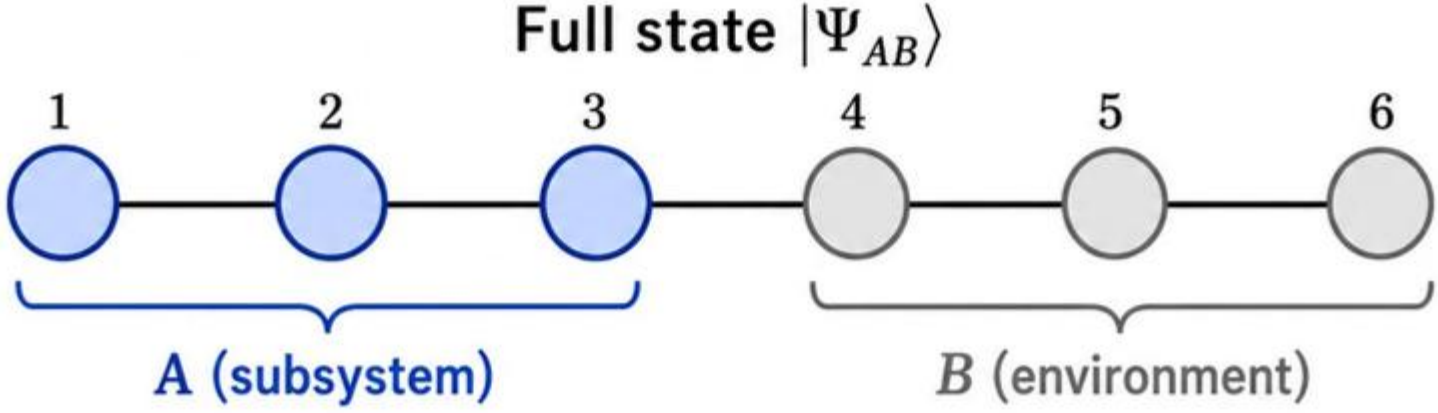
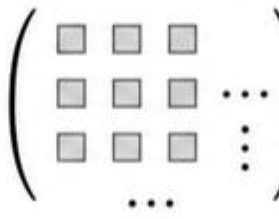
4. Eigenstate → Reduced Density Matrix (RDM)



$$\rho_{AB} = |\Psi_{AB}\rangle\langle\Psi_{AB}|$$



trace out B

$$\rho_A = \mathrm{Tr}_B(\rho_{AB})$$



★ Even if the full state is pure, the subsystem state can be mixed.